Using Design of Experiments: How to create effective Validations and Qualifications

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“A state of statistical control is not a natural state for a manufacturing process; it is instead an achievement, arrived at by eliminating one by one, by determined effort, the special causes of excessive variation."

— W. Edwards Deming

About the presentation

- Presenter is no statistician or statistical process control expert.
- Presenter has worked primarily for MD manufacturing, not pharma or development. – May not be able to address very specific items on these.
- Theoretical approach – showing which tools may be used, but now how to use the tools.
- Could've talked about this forever! Summary of main elements and ideas.
- Questions or input anytime

Terminology

- Design of Experiments:
  - Is the design of any information-gathering exercises where variation is present, under the control of the experimenter.
  - It is a general term that encompasses screening experiments, response surface studies, and analysis of variance.
  - In general, a designed experiment involves purposely changing one or more inputs and measuring resulting effect on one or more outputs.
- ANOVA (Analysis of Variance):
  - A statistical tool that allows to measure the significance of a factor or treatment over a response variable.

Terminology

- Validation:
  - Means confirmation by examination and provision of objective evidence that the particular requirements for a specific intended use can be consistently fulfilled. (820.3 z)
- Verification (or usually Qualification):
  - Means confirmation by examination and provision of objective evidence that specified requirements have been fulfilled. (820.3 aa)

What is variation?

- Variation (or variability) is everywhere, all the time, in everything, it is the “natural state”.
- Variation is the deviation you have from an ideal or required state.
- Divided in two parts:
  - Common Cause (Natural, Chance Cause) Variation → This is expected to be in a process. It requires a team approach towards identifying ways of improving the process.
  - Special Cause → Unusually good or bad trend in a process. This is not expected in a process and requires investigation and correction.

Possible sources of variation – the M's

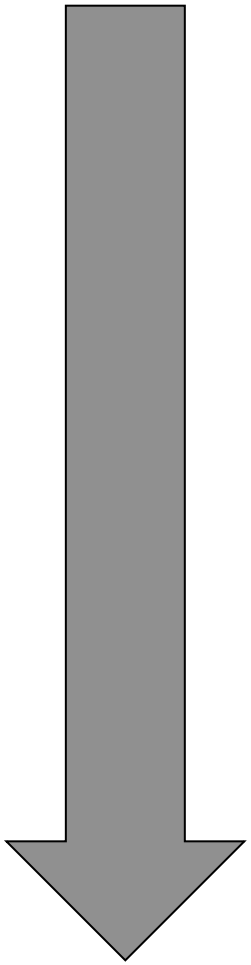
- Instruments → Measurement
- Equipment → Machine
- Facilities and Utilities →Machine
- Processes → Man
- Procedures → Method
- Computer Systems / Software → Machine
- Product Design Changes → Material / Method
- Raw Material → Material
- Temperature, Humidity, Light → "Mother Nature" (Environment)

Possible sources of variation – the M's

- Vibration→ Machine / "Mother Nature"
- Purity of process materials → Material
- Human Factors (training, ergonomic factors, stress, etc) →Man
- Wear of equipment → Machine

Components of Variation in a Process

- Total Process Variation
- Lot to Lot Variation
- Within Lot Variation
- Stream to Stream Variation
- Within Stream Variation
- Time to Time Variation
- Within Time Variation
- Part to Part Variation
- Within Part Variation
- Measurement Error
- Inherent Process Variation



What could be considered an effective Validation or Qualification?

- Put in simple terms, is being able to achieve the requirement of obtaining objective evidence that the requirement can be consistently fulfilled.
- Easier said than done …
  - This implies having a sufficiently good level of process understanding.
  - The use of a lifecycle approach towards validation.
  - Good project management and good archiving that capture scientific knowledge.
  - And, robust statistical analysis of the of the development and the result of the process.

What could be considered an effective Validation or Qualification?

- Validation is then an exercise of confirming through objective evidence that you have understood and controlled the variation of your process.
- You can do variation control in two ways:
  - Reactive: final product inspection / testing.
  - Proactive: QbD, Process Pokayoke → This is where DOE makes a significant contribution to the validation effort effectiveness.

What does FDA considers to be process understanding?

- A good level of process understanding implies that the manufacturer:
  - Understands the sources of variation
  - Detects the presence and degree of variation
  - Understands the impact of variation on the process and ultimately on product attributes
  - Controls the variation in a manner commensurate with the risk it represents to the process and product

Current Process Validation is being performed in 3 phases

- Phase 1: Process Knowledge and Understanding
  - Definition of CQAs (CTQ) and CPPs
  - Characterization of risk
  - Characterization of variability
  - Define the measures for variability mitigation and process control
- Phase 2: Process Qualification
  - Equipment Qualification
  - Product Performance Qualification
- Phase 3: Continued Process Verification
  - Systems for detecting unplanned departures from the process
  - ongoing program to collect and analyze product and process data that relate to product quality

When to apply DOE?

- Simple answer: During Phase 1.
  - The DOE is a process development activity used as the means to reduce the variation related to either the input or output of the process.
- Phase 2? → too late! The process is already designed and you're testing that the process is capable and stable.
  - Other statistical tools like capability analysis or control charting apply.

What type of DOE to apply?

- Factorial
  - Used to identify the degree of contribution of factors to a response variable.
  - Can be either full or fractional.
- Response Surface
  - Used to optimize the interactions of the factors and identify the optimal level of all factors simultaneously.
- Mixture
  - Used when applying DOE to formulations or mixtures.
- Taguchi
  - Simultaneous optimization of factors for both the main effects and noise elements. Also known as robust parameter design.
- Custom

What to consider before DOE?

- Measurement System Analysis! → Before you go ahead and take measurements, make sure you've got validated tools to do it.
  - Use Gage R&R to demonstrate the ability of the equipment to provide accurate measurements.
  - Calibration of equipment
  - Select personnel to perform the tests – trained and capable personnel.
  - Analyze the response variable with other statistical tools: normality and skewness.

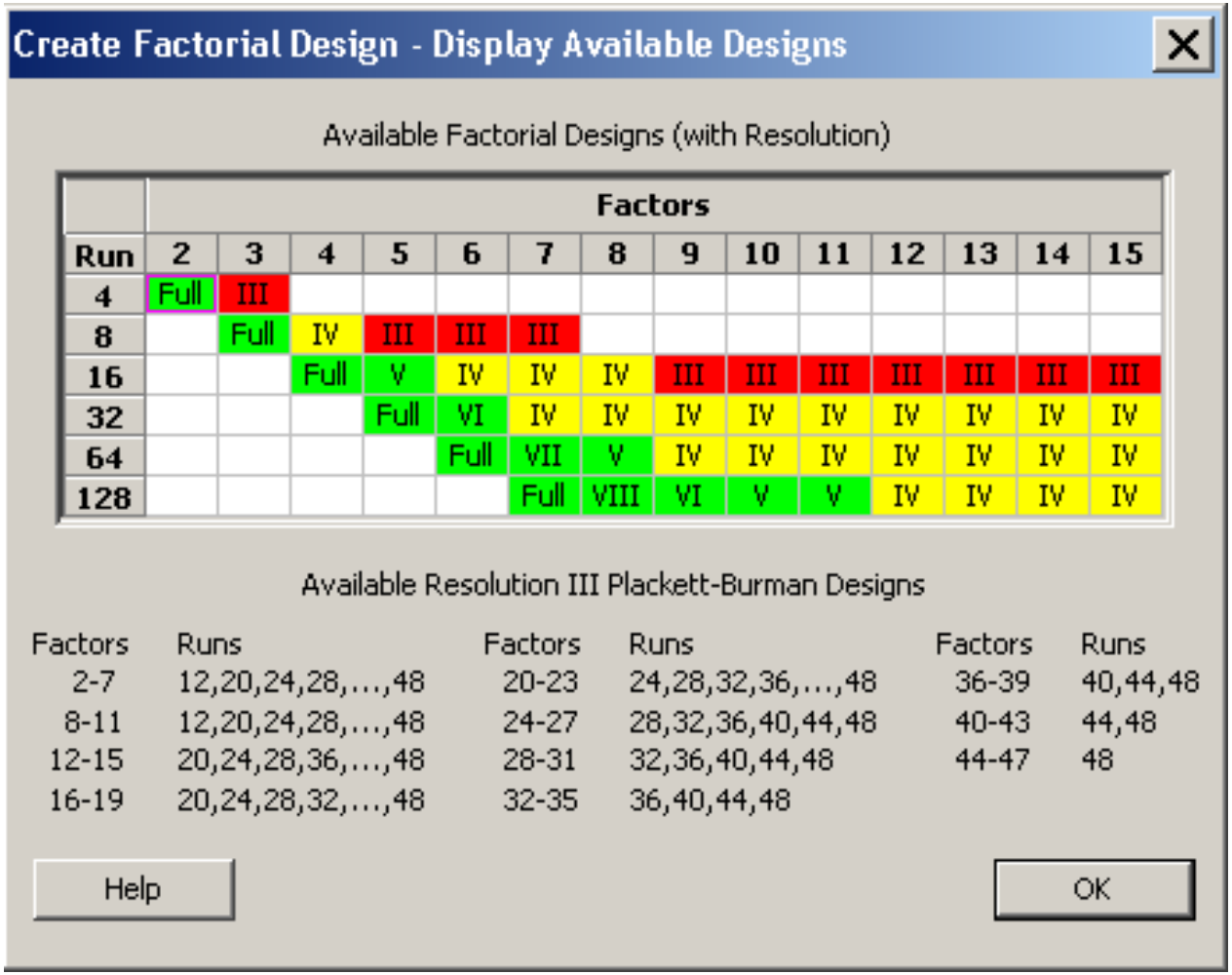
How to apply DOE?

- Step 1: Identify a group of variables that may impact the response variable.
  - This is usually a multidisciplinary team activity.
  - SMEs from the process should perform brainstorming for this.
  - Based on scientific rationale.
  - Output (response) variables may come out of a HHA (Health Hazard Analysis) or FMEA (Failure Modes and Effects Analysis) exercise where the safety related elements of the product may be extracted.

How to apply DOE?

- Step 2: Determine the number of runs and replicates to perform with the study.
  - Number of runs based on level of detail that the results should provide.
  - Commensurate with the risk of the process.
  - May be an iterative approach:
    - May start with screening designs for confirming the main factors.
    - Then run full factorials over the few main contributors to fully understand their impact on variation (interactions).

How to apply DOE?



Taken from MINITAB® Version 15

How to apply DOE?

- Step 3: Measure the output variable for all combinations of the input factors.
  - Measurements should be taken whenever possible in a random order to avoid bias.
  - Replications may be taken for additional certainty on the results of the impacted variability.

How to apply DOE?

StdOrder	RunOrder	CenterPt	Blocks	Temperature	Pressure	Catalyst	Yield
2	1	1	1	40	1	A	66
14	2	1	1	40	1	B	66
4	3	1	1	40	4	A	102
3	4	1	1	20	4	A	98
1	5	1	1	20	1	A	65
16	6	1	1	40	4	B	54
11	7	1	1	20	4	A	107
7	8	1	1	20	4	B	68
6	9	1	1	40	1	B	53
8	10	1	1	40	4	B	66
5	11	1	1	20	1	B	55
9	12	1	1	20	1	A	85
12	13	1	1	40	4	A	108
10	14	1	1	40	1	A	89
15	15	1	1	20	4	B	52
13	16	1	1	20	1	B	63

How to apply DOE?

- Step 4: Full variables analysis.
  - The idea is to extract the significant factors.
  - F function analysis, normality test and pareto charts are used to segregate the significant factors.
  - Main contributors may include not only main factors but also interactions.

How to apply DOE?

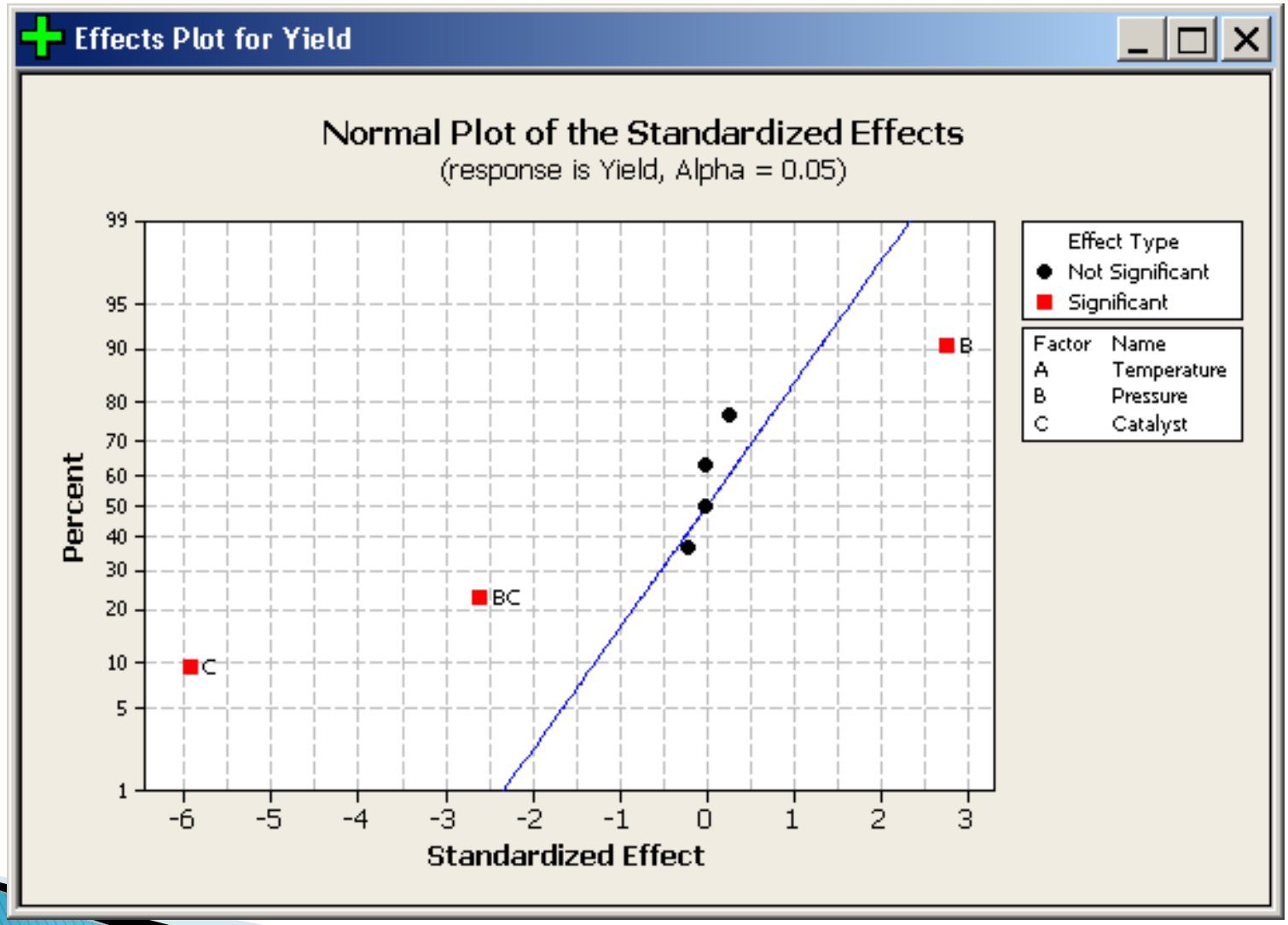
Factorial Fit: Yield versus Temperature, Pressure, Catalyst

Estimated Effects and Coefficients for Yield (coded units)

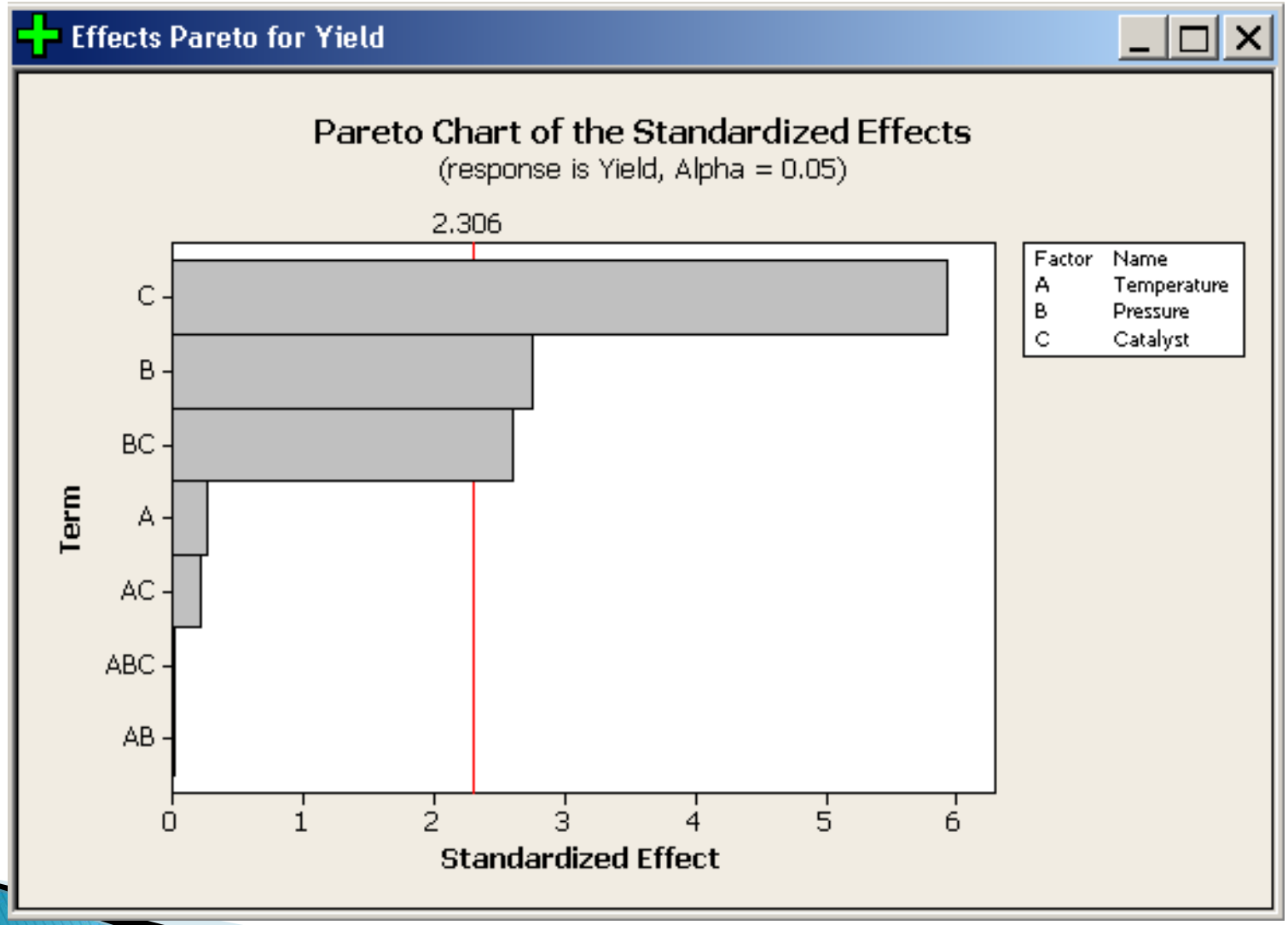
Term	Effect	Coef	SE Coef	T	P
Constant		74.81	2.561	29.21	0.000
Temperature	1.38	0.69	2.561	0.27	0.795
Pressure	14.12	7.06	2.561	2.76	0.025
Catalyst	-30.38	-15.19	2.561	-5.93	0.000
Temperature*Pressure	-0.13	-0.06	2.561	-0.02	0.981
Temperature*Catalyst	-1.13	-0.56	2.561	-0.22	0.832
Pressure*Catalyst	-13.37	-6.69	2.561	-2.61	0.031
Temperature*Pressure*Catalyst	-0.13	-0.06	2.561	-0.02	0.981

S = 10.2439 PRESS = 3358
R-Sq = 86.14% R-Sq(pred) = 44.55% R-Sq(adj) = 74.01%

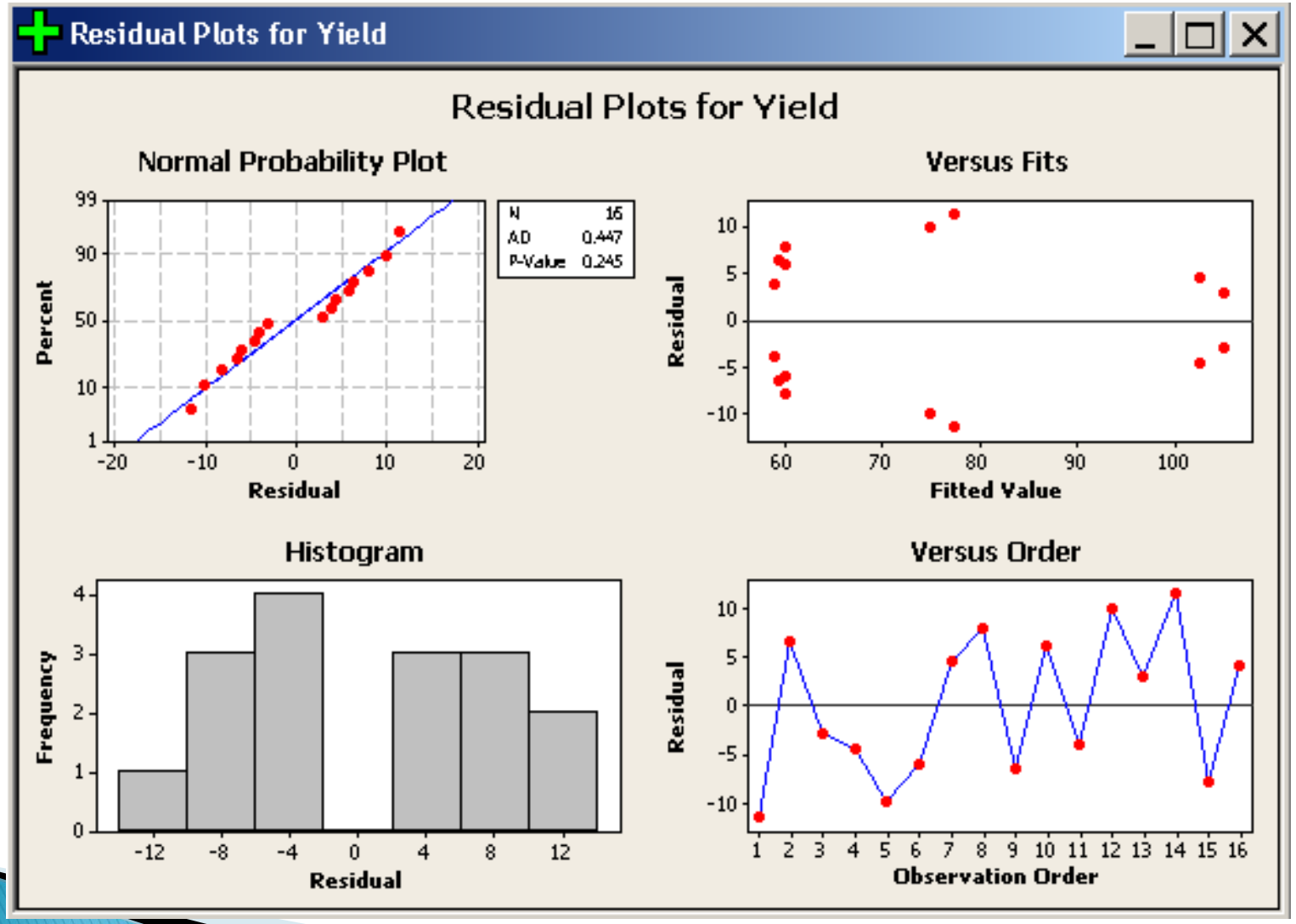
How to apply DOE?



How to apply DOE?



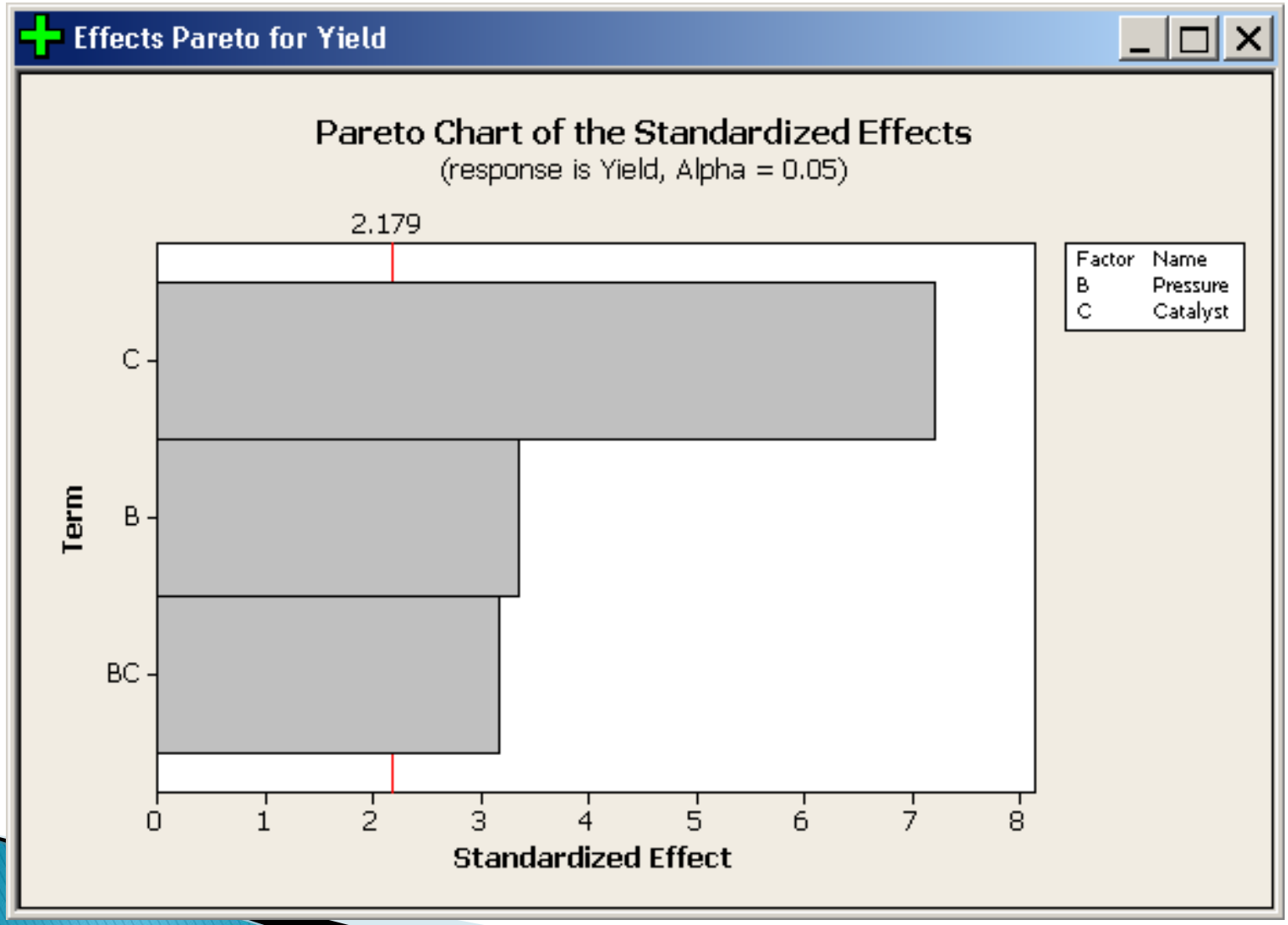
How to apply DOE?



How to apply DOE?

- Step 5: Screening and Optimization analysis
  - Focus on the statistically significant factors only.
  - This will provide additional details about the actual behavior of the response variable due to the significant effects only and reduce noise to the analysis.

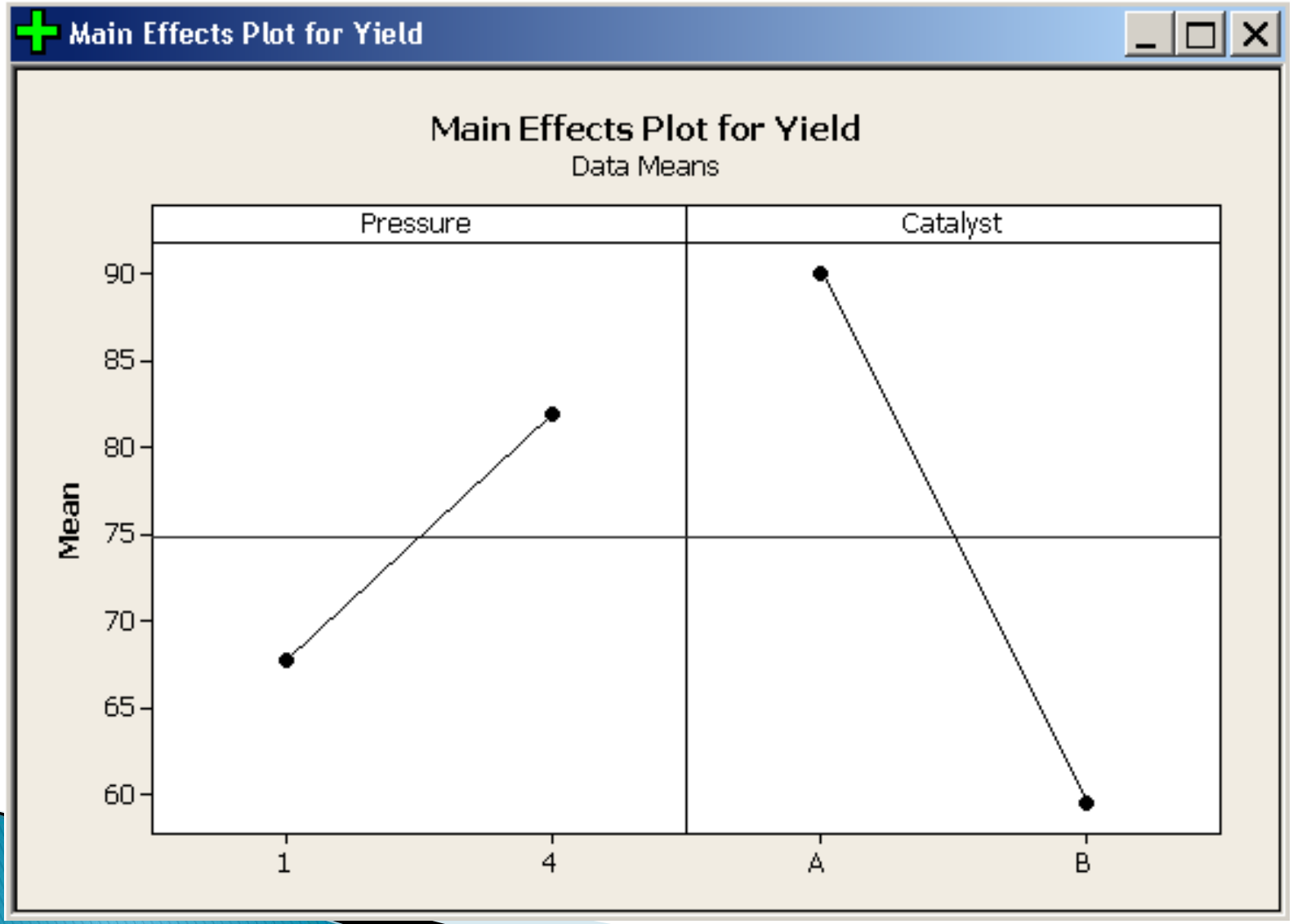
How to apply DOE?



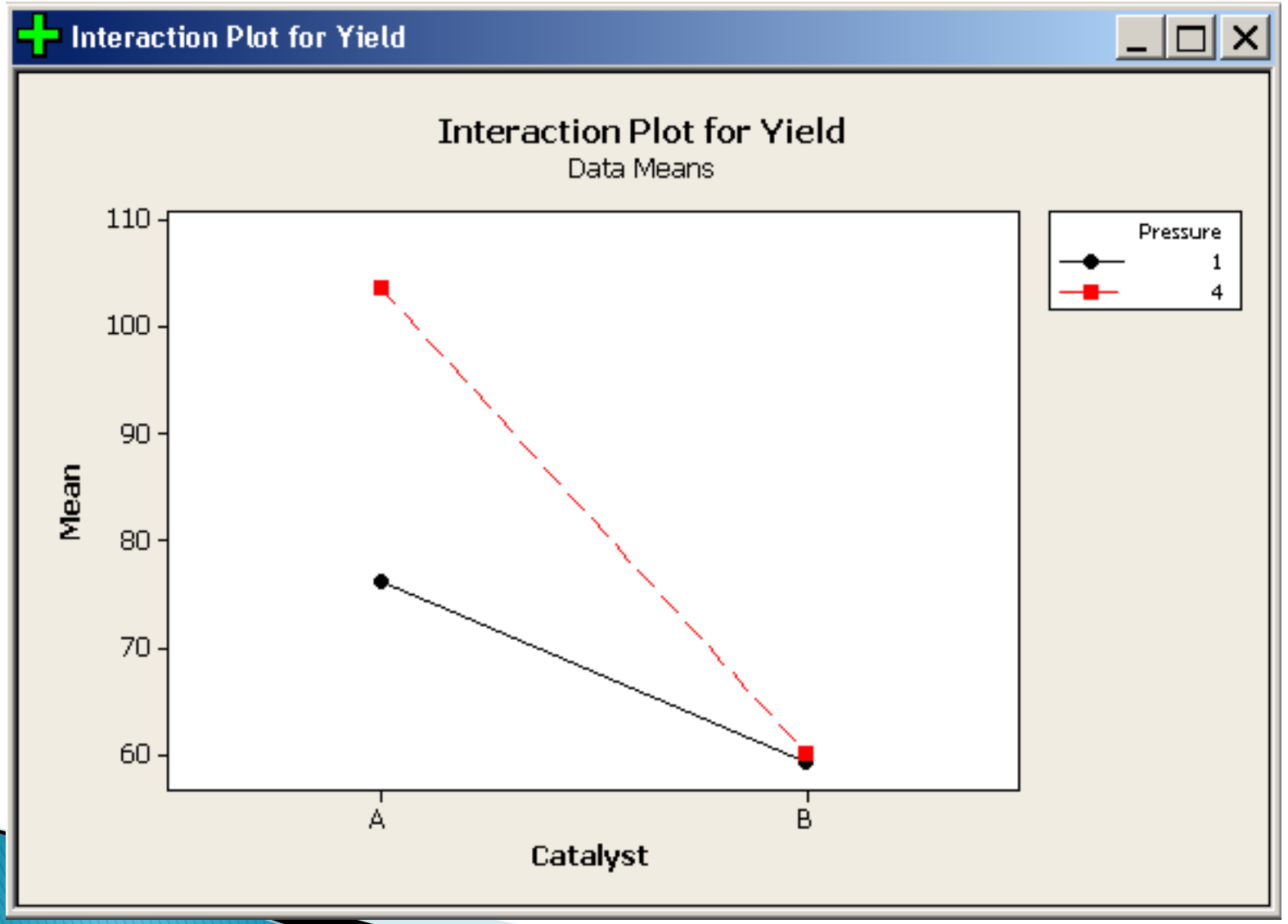
How to apply DOE?

- Step 6: Conclusions and decision making
  - The idea is to optimize the experiment to draw the required conclusions about the process.
  - F function analysis, normality test and pareto charts are used to segregate the significant factors.
  - Main contributors may include not only main factors but also interactions.

How to apply DOE?



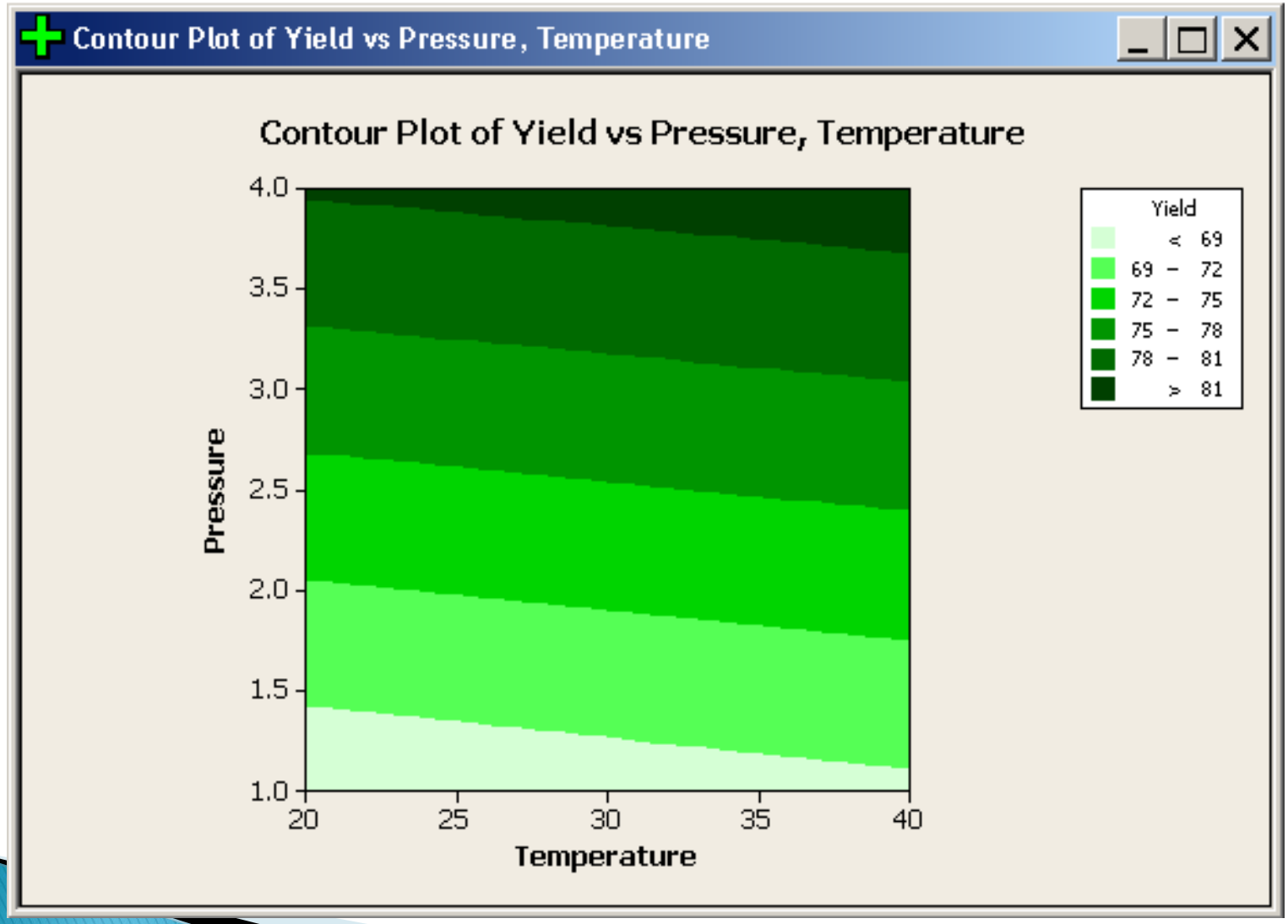
How to apply DOE?



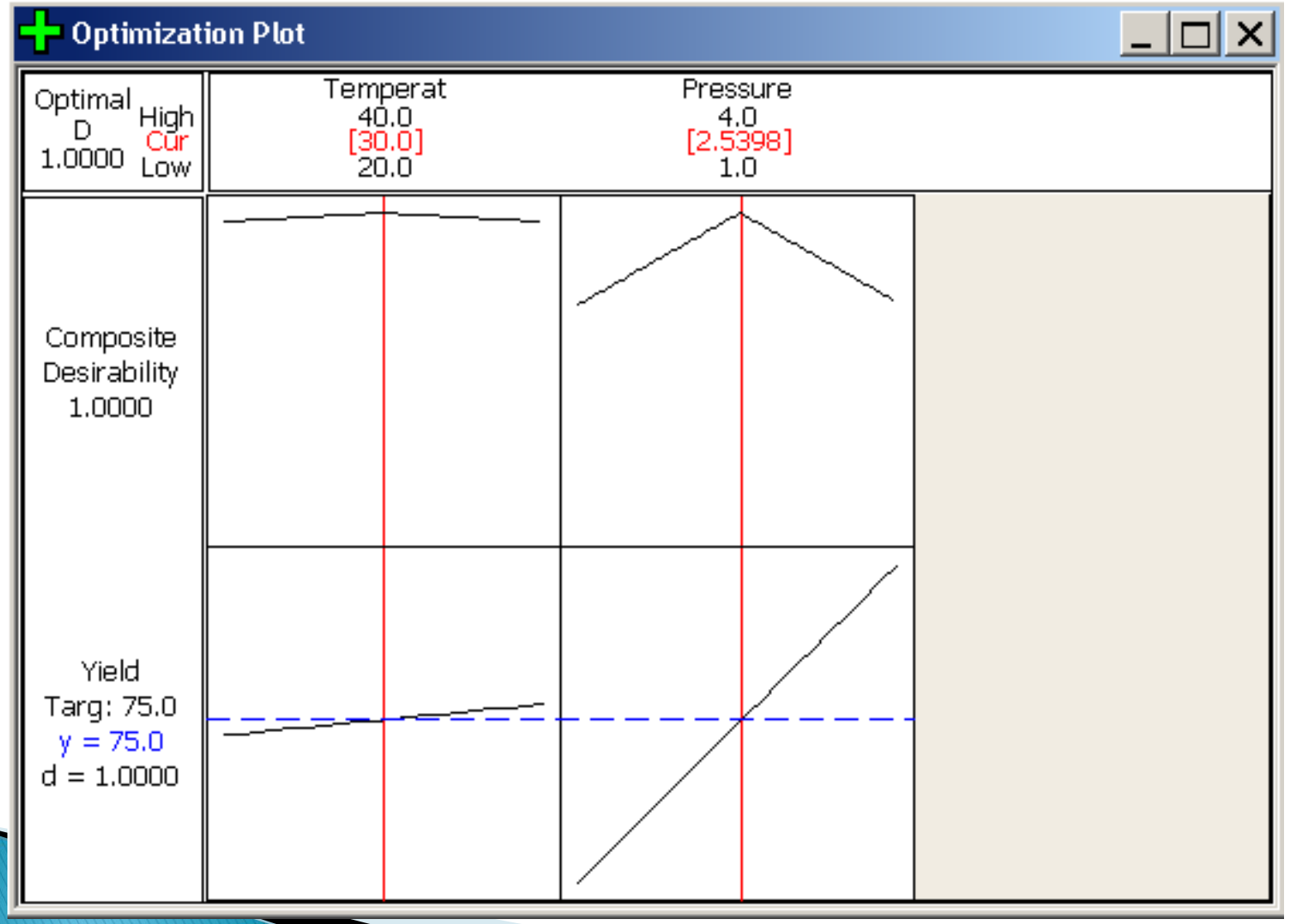
What results from the DOE?

- Some conclusions that will feed into the validation exercise:
  - *The Critical Process Parameters that impact Yield are the Catalyst, the Pressure, and their interaction.*
  - *There are additional variables to be accounted for the variability of the Yield that are not explained by the model.*
  - *The best yield is obtained with Catalyst A at 4 pounds of pressure.*

Response Surface Examples



Response Surface Examples



What else to do after DOE?

- Perform additional testing to confirm results.
- Try other values of the factors to identify the optimal operation point, nominal operation.
- Identify potentially missing variables from the original brainstorming.
- Perform DOE studies for all variables responsible for CQAs.

Tools for PPQ execution

- Use capability analysis, challenge tests or acceptance plans as applicable to perform the testing required during PPQ.
  - Capability Studies measure the ability of the process to consistently meet the specifications. It is appropriate for measurable characteristics where nonconformities are due to variation and off-target conditions.
  - Challenge tests demonstrate that mistake proofing methods designed to detect or prevent errors are working.
  - Acceptance Plans optimize the number of samples to be tested and aids in demonstrating conformance to specification.

Focus on the Summary Report

- The summary report is most usually the presentation card for auditors in regards to the validation activity performed.
- A good summary report will prevent the auditor from deepening into the actual data or protocol results.
- Report should be easy to read and understand, but be complete in its content.

A word on PAT

- Process Analytical Technology – Solution offered by FDA for pharma to perform QbD
  - Reducing production cycle times by using on-, in-, and/or at-line measurements and controls
  - Preventing rejects, scrap, and re-processing
  - Real time release
  - Increasing automation to improve operator safety and reduce human errors
  - Improving energy and material use and increasing capacity
  - Facilitating continuous processing to improve efficiency and manage variability

What to do with Legacy?

- Manufacturers of legacy products can take advantage of the knowledge gained from the original process development and qualification work as well as manufacturing experience to continually improve their processes. → Start in Phase 3: Continued Process Verification and go backwards.
- Risk based approach
- Remediation plans
  - Can't do it immediately, but over time in a planned manner.
- Continuous Improvement

Interactive Exercise

- Instructions: Read through the Validation example provided and answer the following questions:
  - Which statistical tools could be identified in the example validation?
  - The Validation Plan describes the process validation approach that was used, where do you see DOE come into play in the process?
  - What differences could you point out from what you've experienced that is currently being done?
  - What do you think is needed to generate validation documentation that will be similar to the example provided?
- Time for review: 15 minutes, then 10 minute discussion

Thank you